

## Practical Assignment 1

### Analysis of Data Sources

As an **example scenario** for all practical tasks, we will consider the beverage market chain *DrinkMix* which sells various drinks such as beer, wine, soda, etc. to customers in each of its branches. Recently, the company fused with some regional competing companies, which resulted in a heterogeneous IT infrastructure. In it, there are several small transactional databases, which manage the sales and customer data of the six individual stores. These databases are used by independent applications. A planned data warehouse system, which will draw its data directly from the distributed store databases, is now supposed to help analyzing customer buying behavior. Company management has specified which analyses are to be possible with the system, how they are to be prepared and in what form they are to be processed. As the key business figure, product sales should be the focus. The essential evaluation and observation perspectives will be presented in the second assignment.

Throughout your work on all assignments, the analysis of the data sources, the creation of the data warehouse and the implementation of the ETL process will essentially be recreated. Programming of an explicit analysis tool will not be part of the practical tasks. From a technological point of view, the DBMS Oracle, as well as the languages Oracle SQL and Oracle PL/SQL will be used. The access data for the respective database account will be sent to each group individually by e-mail. Instructions on how to connect to the DBS can be found on the Moodle page. Out of the six existing branches of the company, three branches will be assigned to each group for the entire practical course. On Moodle, you can find the tool *Oracle SQL Developer*, which you may use to connect to the DBS. Alternatively, the tasks can also be solved with other development environments like DBeaver or DataGrip<sup>1</sup>.

The organization of the data is as follows:

- Every student group is assigned a database user, i. e. there is the user DWT\_GROUP\_01 for group 1, etc. This requires that you register into a group on Moodle first.
- Each student group may create, delete and modify tables and data in their own area as they wish.
- To simplify things, in this training scenario the six branches of the company are not in different databases, but exist as database users in the DBS, i. e. there are users DWT\_BRANCH\_1 to DWT\_BRANCH\_6
- Every student group can only read data of the branches that were assigned to them.

To pass the practical part of the course, you must complete and pass all of the assigned practical task sheets.

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<sup>1</sup>Note that we cannot provide support for such alternative software.

## Tasks

This part of the project focuses on accessing and analyzing data sources. To successfully complete this first assignment, you will need to complete all of the following points:

1. Use the username and password you received via email to connect to the DBS! Familiarize yourself with working on the database!
2. The table structures of the individual store databases are initially unknown. Your task is to determine the underlying physical database schema (attributes, data types, integrity conditions, etc.) of each of the branches you were assigned. Use targeted queries (SELECT statements) to system tables of the DBMS Oracle to perform this task. Write down the used queries!
3. Write the corresponding DDL statements for each database schema of your branches! (In other words, the CREATE statements that were used to create the tables of your branches.)
4. Based on the observed schemas and DDL statements, draw an E/R schema for each of your three assigned branches!  
Important: Do not guess relationships, entities or attributes! Only the elements in the E/R schema that are explicitly present in the corresponding database schema are to be modeled! If there are any ambiguities in the mapping from CREATE statement to E/R model, please record them in writing.
5. The schemas/tables of the branches contain various *problems*, such as as poor design. Find as many as possible and make a list of them for each of the branches.

Submit your results in the appropriate format via the Moodle submission page until the specified deadline! With regard to this, please also refer to our deadline and plagiarism policies. Good luck with your work!

## Additional Requirements

Please also keep these requirements in mind when submitting your solution:

- You must submit your results in the correct format. SQL code should be submitted in a .sql file, whereas text and diagrams should be submitted as .pdf only! Multiple files should be bundled in a .zip file.
- Do not submit hand-drawn diagrams, ASCII art, screenshots or similar low-effort solutions! To create the E/R schemas, use graphical modeling tools such as Dia, Visio, LibreOffice Draw, Inkscape, draw.io, etc.
- E/R diagrams should follow the notation described by *Elmasri and Navathe*<sup>2</sup>. At the very least, diagrams must adhere to *Chen's notation*<sup>3</sup> for Entity-Relationship modeling.

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<sup>2</sup>Ramez Elmasri and Shamkant B. Navathe, *Fundamentals of Database Systems*, 7th Edition, Pearson, 2015.

<sup>3</sup>Peter P. Chen, "The Entity-Relationship Model: Toward a Unified View of Data," *ACM Transactions on Database Systems (TODS)*, vol. 1, no. 1, pp. 9–36, 1976.